of θ are *also* plausible with respect to E; in topological terms, this means precisely that there is an open set $U \in \mathcal{O}(\Theta)$ such that if $\theta' \in U$ then θ' is "sufficiently close" to θ.[4]

Under this condition, we can give a sufficient condition for when a null hypothesis H_0 is nonconfirmable:

Proposition 3.3. Let Θ be a topological space and c a open-valued non-refuting confidence function. Suppose that H_0 has empty interior. Then for every w, $w \Vdash \Diamond H_1$; i.e. $w \Vdash \neg\Box H_0$.

In particular, for a two-sided hypothesis $H_0 : \theta = \theta_0$, no world w satisfies $w \Vdash \Box H_0$.

Proof. Since H_0 has empty interior, H_1 is dense. Since c is open-valued non-refuting, $c(E_w)$ is an open nonempty set for all E_w. Thus $c(E_w) \cap H_1 \neq \varnothing$. Thus $w \Vdash \Diamond H_1$; equivalently, $w \Vdash \neg\Box H_0$. □

This observation extends to a characterization of the realizability of $\Box H_i$ and $\Diamond H_i$ by possible worlds.

Theorem 3.4. Let Θ be a connected topological space, c an open-valued confidence function, Ω the sample space.

Suppose further that the image $c(\Omega^{<\omega}) \subset 2^{\Theta}$ satisfies

(1) (Θ-Exhaustiveness) for each $\theta \in \Theta$ there exists an $E_\theta \in \Omega$ such that $\theta \in c(E_\theta)$.
(2) (Precision) For each open $U \in \mathcal{O}(\Theta)$ there is evidence $E_U \in \Omega^{<\omega}$ such that $c(E_U)$ is nonempty and $c(E_U) \subseteq U$.

Suppose that $H \subseteq \Theta$ is nonempty. Then

(1) (Satisfiability of $\Box H$) There is a world w such that $w \Vdash \Box H$ if and only if H has nonempty interior, and
(2) (Satisfiability of $\Diamond H \wedge \Diamond H^c$) There is a world such that $w \Vdash \Diamond H \wedge \Diamond H^c$ if and only if H^c is nonempty.

Proof. First, let H have nonempty interior H°. By precision of c, there is some E_H such that $c(E_H) \subseteq H^\circ \subseteq H$. Select any $\theta \in \Theta$ and let $w = (\theta, E_H)$. Then $w \Vdash \Box H$. Conversely, suppose that H has empty interior. Then by 3.3, for every w we have $w \Vdash \Diamond\neg H$, so $w \Vdash \neg\Box H$. Thus $\Box H$ is not satisfiable.

Hence, $\Box H$ is satisfiable if and only if H has nonempty interior.

Finally, since H and H^c are nonempty and Θ is connected, the boundary $\partial H = \partial H^c$ is nonempty. Let $\theta_H \in \partial H$. By assumption on c, there exists E_{θ_H} such that $\theta_H \in c(E_{\theta_H})$. Since $\theta_H \in \partial H$, any open set containing θ_H intersects H and H^c nontrivially. Hence $H \cap c(E_{\theta_H})$ and $H^c \cap c(E_{\theta_H})$ are nonempty. Thus the world $w = (\theta_H, c(E_{\theta_H}))$ has $w \Vdash \Diamond H \wedge \Diamond H^c$. □

The conditions on confidence intervals in Theorem 3.4 are satisfied by two-sided Wald confidence intervals, but not by *one-sided* confidence intervals. One-sided confidence intervals are insufficient because their length is always infinite, but precision requires the confidence interval to be containable within arbitrary open sets of finite length given an appropriate stream of finite data.

This topological criterion immediately highlights the difference between two-sided hypothesis testing versus the two one-sided hypothesis testing paradigms.

Corollary 3.5. Let Θ, c, Ω be as in 3.4 be a topological space with no open points.[5] Then

(1) If $H_0 = \{\theta\}$, then $\Box H_1$ and $\Diamond H_0 \wedge \Diamond H_1$ are satisfiable, but not $\Box H_0$.

[4] It is worth reviewing a few subtleties pertaining to the use of open sets. Regarding discrete sets of parameters Θ, we may assign Θ the discrete topology. In this case, every subset of Θ is both closed and open. Regarding the use of open-valued confidence sets, one might argue that in the construction of confidence intervals via the bootstrap, the confidence interval should be a *closed* interval with nonempty interior so that the confidence interval nominally contains the $1-\alpha$ endpoints. To assuage this concern, I note that if one replaces "open-valued" with the constraint that $c(E)$ is the closure of a nonempty open set, all subsequent results in this section remain true as the proofs require only that $c(E)$ have nonempty interior.

[5] Containing no open points is entailed in the case that Θ is connected and not a single point by stronger separation axioms, such as Θ being Fréchet (T1) or Hausdorff (T2).

(2) If H_0, H_1 is such that the interiors H_0° and H_1° are both nonempty, then $\Box H_0$, $\Box H_1$, and $\Diamond H_0 \wedge \Diamond H_1$ are all satisfiable.

Proof. By 3.1, for each world exactly one of $\Box H_0$, $\Box H_1$, or $\Diamond H_0 \wedge \Diamond H_1$ is true. The rest follows immediately from 3.4. □

To assert that a true effect size is *exactly* zero is to posit a hypothesis with nonempty interior, namely, $H_0 = \{\theta\}$ inside a space Θ where no point is open. By contrast, if $H_0 = [-M, M]$ with $M \in (0, \infty)$ then H_0 and H_1 are both confirmable.

4. Revisiting Type I and Type II Errors

Recall the distinction between a Type I and Type II error from hypothesis testing:

(1) A type I error occurs when we reject H_0 despite H_0 being true. In our formal symbolism: $\Box H_1 \wedge H_0$.
(2) A type II error occurs when we fail to reject H_0 when H_1 is true; i.e. $\Diamond H_0 \wedge H_1$.

Considering the situation in a type II error an *error* seems to hinge upon the understanding of hypothesis testing as a *default logic*: type II errors occur when one retains H_0. However, if one adopts a modal language, inferring the *possibility* of H_0 in light of the test and accrued evidence E is less an error in the conclusion than the failure of the test to result in a decisive outcome.

To this end, the modally-inspired analogues of Type I and Type II errors for the logic of Two One-Sided Testing are given as follows:

(1) A *decisive error* occurs when $\Vdash \Box H_i \wedge H_{1-i}$, and
(2) *Indecisiveness* occurs when $\Vdash \Diamond H_i \wedge \neg\Box H_i$; or, equivalently, $\Vdash \Diamond H_0 \wedge \Diamond H_1$.

We assume throughout this section that the events $\Box H_i$ and $\Diamond H_i$ are all measurable events for each $\theta \in \Theta$, and that two-sided $1-\alpha$ confidence intervals exist for Θ. Note that we may think of the confidence interval $c(E)$ as a random set as a function of the evidence E.

4.1. Decisive Error.

The rate of decisive errors is closely related to the coverage of the confidence interval.

Proposition 4.1. Let c be a $(1-\alpha)$ confidence region. Then the d-value (”d” for ”Decisive”) of the test,

$$d_\Box = \max\Big(\sup_{\theta\in\Theta_0} \mathbb{P}_\theta(\Box H_1), \sup_{\theta\in\Theta_1} \mathbb{P}_\theta(\Box H_0)\Big)$$

satsifies

$$d_\Box \le \alpha.$$

Proof. For a given $\theta \in H_0$ and level $1-\alpha$

$$\begin{aligned} \mathbb{P}_\theta\big(\Box H_1\big) &= \mathbb{P}_\theta\big(c(E) \subseteq H_1\big) \\ &\le \mathbb{P}_\theta\big(c(E) \cap \{\theta\} = \varnothing\big) && \text{(Since } \theta \notin H_1\text{)} \\ &\le \alpha && \text{(By definition of confidence interval)} \end{aligned} \tag{15}$$

Replacing H_0 with H_1 we can conclude that for $\theta \in H_1$, $\mathbb{P}_\theta\big(\Box H_0\big) \le \alpha$. Thus

$$\max\Big(\sup_{\theta\in\Theta_0} \mathbb{P}_\theta(\Box H_1), \sup_{\theta\in\Theta_1} \mathbb{P}_\theta(\Box H_0)\Big) \le \max(\alpha, \alpha) = \alpha. \tag{16}$$

as desired. □

Note that the bound of α may not be sharp: in TOST, for example, a two-sided $(1-2\alpha)$ confidence interval achieves significance α for testing $H_0 : |\theta - \theta_0| \ge M$ [5].

4.2. **Power Analysis and Indecisiveness.** Recall that the power function measures the probability of rejecting the null H_0, assuming the true parameter is $\theta \in H_1$[6] and $E \in \Omega^n$ is drawn from the distribution $\theta^{\otimes n}$. In other words,

$$\beta(\theta, n) = \mathbb{P}_\theta(c(E) \cap H_0 = \varnothing) = \mathbb{P}_\theta(\Box H_1).^7 \tag{17}$$

We can extend this to define the *decisive power function*, which make symmetric the roles of H_0 and H_1 and measure how frequently the test will yield a decisive outcome of $\Box H_0$ or $\Box H_1$:

Definition 4.2. The *partial decisive power functions* δ_0, δ_1, are given by

$$\begin{array}{rcl} \delta_0(\theta, n) & = & \mathbb{P}_\theta(\Box H_1), \\ \delta_1(\theta, n) & = & \mathbb{P}_\theta(\Box H_0). \end{array} \tag{18}$$

The *decisive* power function $\delta(\theta, n)$ is the probability of reaching a decisive result, given θ:

$$\begin{array}{rcll} \delta(\theta, n) & = & \mathbb{P}_\theta(\Box H_0 \vee \Box H_1) & \\ & = & \mathbb{P}_\theta(\Box H_0) + \mathbb{P}_\theta(\Box H_1) & \text{(By disjointness of the events } \Box H_0 \text{ and } \Box H_1) \\ & = & \delta_1(\theta, n) + \delta_0(\theta, n). & \end{array} \tag{19}$$

Observe that for a given null hypothesis H_0, the decisive power function $\delta(\theta, n)$ is always at least as large as the usual power function $\beta_0(\theta, n)$:

Proposition 4.3. Let H_0 be a hypothesis, δ its decisive power function, and β its power function. Then the decisive power function $\delta(\theta, n) \geq \beta(\theta, n)$.

Proof. Since $\delta_0 = \beta$ is the power function for H_0 and both partial decisive power functions δ_i take values in $[0, 1]$, $\delta(\theta, n) \geq \beta(\theta, n)$. ☐

This signals that that the power function may fail to capture all of the confirmation-theoretic information encapsulated by a given test. We show a case where this can happen, and one where it doesn't.

Example 4.4. (Decisive Power Equals Power for Two-Sided Tests) Let $H_0 = \{\theta\}$ and $\delta(\theta, n)$ its decisive power function. Let $\theta \in H_1$. As we have already established, $\Vdash \neg H_0$, so $\mathbb{P}_\theta(\Box H_0) = 0$. Thus

$$\begin{array}{rcl} \delta(\theta, n) & = & \mathbb{P}_\theta(\Box H_0 \vee \Box H_1) \\ & = & \mathbb{P}_\theta(\Box H_0) + \mathbb{P}_\theta(\Box H_1) \\ & = & 0 + \delta_0(\theta, n) \\ & = & \beta(\theta, n). \end{array} \tag{20}$$

Example 4.5. (Decisive Power Exceeds Power for TOST) Let $H_0 = [-M, M]$ and $\delta(\theta, n)$ its decisive power function. Let $\theta = M$. Then for a sufficiently large n the confidence interval $c(E)$ will be contained inside H_i with positive positive probability for $i = 0, 1$. Thus $\delta_0(M, n), \delta_1(M, n) > 0$ and so $\delta(M, n) > \delta_0(M, n) = \beta(M, n)$.[8]

We now revisit the concept of Type II errors discussed in section 2.1. By contrast with the default account of hypothesis, the confirmatory logic of hypothesis testing can avoid the pitfalls of Type II errors by not imposing normative constraints on an investigator's beliefs in the event of an inconclusive test. On this view, failing to reject a hypothesis, i.e. $\Diamond H_0$, does not impose any doxastic constraints on an agent: one can continue to believe either H_0 or H_1. Unlike the default position, a null hypothesis H_0 need not be justified *a priori*, as the modal approach does not constrain an agent to hold particular beliefs regarding the null. A Type II error, then, is not an error in an agent's doxis, since the result

[6]This is not strictly necessary; the power function makes sense for $\theta \in H_0$ as well. But the "power" $\beta(\theta, n)$ when $\theta \in H_0$ is the probability of rejecting H_0, which in that case is the same as the significance of the test at θ.

[7]Note that the dependence on n is suppressed in this notation: $c(E)$ is a function of E, which in turn is an event of n observations drawn from the true data generating process.

[8]Observe that in this case the decisive power of the test was witnessed by the possibility of a Type I error at the boundary of the TOST interval. However, the rate of false positives is still bounded above by α by the construction of $c(E)$.